

ARK-452 Preregistration

Multi-Step Workflow With One Invalid Step

Series: ExecutionProof ARK Authorization-Boundary Track

Version: v1.0-draft → to be locked as v1.0 immediately before hardware submission

Repository: <https://github.com/derekhone/executionproof-testbeds>

Folder: ark-452/

Target tags: ark-452-v1.0-lock (preregistration LOCK) → ark-452-v1.0 (execution result)

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Date drafted: 2026-07-17

Lock date: To be set at moment of hardware submission

Prerequisite: ARK-449 should be executed and published before ARK-452 is submitted. ARK-449 establishes the single-step state-change boundary. ARK-452 extends it to sequential multi-step chains. If IBM budget requires sequencing, ARK-449 takes priority.

Pre-Lock Drafting Correction Notice. During drafting (before any lock, commit, or hardware submission), the mandatory noiseless dry-run caught one construction defect and it was corrected in place: the ALLOW **execution gate was H**, corrected to **X**. A Hadamard would have put the payload in a 50/50 superposition ($P(Q_P=1) \approx 0.50$), making criterion C1 ($S_A_{\min} \geq 0.90$) mathematically impossible — a guaranteed FAIL by construction. The execution gate now drives the payload to $|1\rangle$ via **X**, giving $P(Q_P=1) \approx 1.00$ on a clean device, consistent with the ALLOW/DENY semantics used throughout the corpus (mirrors the ARK-449 v1.1 correction). The SPAM_P circuit retains **H** by design — it needs the $|+\rangle$ symmetry diagnostic. Post-correction noiseless dry-run: ALLOW arms = 1.0000, all seven DENY arms = 0.0000, SPAM gate passed. No criteria, hypotheses, arms, or thresholds are affected; this notice is recorded for full transparency and is locked together with the corrected circuit.

LOCK INSTRUCTION

Before any hardware job is submitted, the following steps must be completed in strict order:

1. Finalize this document. No further changes after step 2.
2. Compute SHA-256 hash of this file and all circuit code files.
3. Commit the MANIFEST (see Section 16) to `executionproof-testbeds`.
4. Push and tag the commit `ark-452-v1.0-lock`.
5. Submit the SPAM gate job. Record the job ID in the execution log **before any results are read**.
6. Submit the principal job. Record the job ID in the execution log **before reading SPAM results**.
7. Read results only after both job IDs are committed.

No criterion may be changed, loosened, or added after step 2. A FAIL is a valid, publishable result. An ABORT at the SPAM gate is a valid, publishable result.

1. Preamble and Series Context

1.1 Where This Fits

The ARK series through ARK-449 (when executed) establishes:

Experiment	Dimension Established
ARK-441 / ARK-446	Boundary exists and replicates across devices
ARK-442	Temporal: stale authority fails closed
ARK-444	Integrity: altered decision record fails closed
ARK-443	Multi-party: single compromised authorizer cannot bypass quorum
ARK-445 / ARK-445b	Three-state: HOLD is a metrologically separable first-class outcome
ARK-447	Noise: Pauli twirling provides modest, measurable improvement
ARK-448	Protocol: gate-stop halts before spending budget on marginal conditions
ARK-449	State-change currency: authorization must remain valid at execution time

What the corpus has not yet tested: **whether the control model enforces authorization at the step level within a sequential workflow — specifically, whether one invalid step halts the workflow before an irreversible execution step, regardless of how many valid steps preceded it.**

This is the gap ARK-452 closes.

1.2 The Enterprise Problem Being Modeled

Real AI agents do not perform isolated single actions. They operate through chains: they gather context, compute, request approval, act on a resource, and write a record. Enterprise governance failures in agentic systems routinely arise not from a single unauthorized action but from authorization logic that treats the entire chain as a unit — or worse, allows later steps to inherit authorization from earlier ones.

The failure pattern this experiment tests:

An agent received approval for step 3 (authorize payment). Steps 1 and 2 passed cleanly. Step 4 (execute the transfer — the irreversible step) is now presented. The agent treats the prior passes as sufficient authorization for step 4.

This is exactly wrong. Step 4 requires its own valid authorization. Prior step passes are not a blank check for subsequent steps.

1.3 Precise Differentiation from ARK-449

ARK-449 tested a single-action authorization boundary when world state changed between T_1 and T_3 . The action was one unit with two evaluation points.

ARK-452 tests a sequential multi-step workflow where each step has an independent authorization requirement, and the irreversible execution step (step 4) must be independently authorized — it cannot inherit authorization from preceding steps.

Dimension	ARK-449	ARK-452
Action structure	Single action	Sequential 5-step workflow
Authorization model	One approval at T_1 , one re-check at T_3	Independent per-step authorization
Failure mode tested	World state change between approval and execution	One invalid step in a multi-step chain
Key claim	Authorization currency expires with world state	Prior valid steps $\neq$ blanket authorization for subsequent steps
What “fails closed”	Execution when state has changed	Irreversible step when any step in the chain is inadmissible

1.4 Workflow Modeled

The workflow has five steps:

Step 1: Read Data	(data retrieval – non-destructive)
Step 2: Calculate	(computation – non-destructive)
Step 3: Approve Payment	(authorization decision – non-destructive)
Step 4: Execute Payment	(IRREVERSIBLE – funds leave the account)
Step 5: Write Record	(audit log – non-destructive post-execution)

Step 4 is the irreversible execution step. The experiment’s central question is whether the control model blocks step 4 when any one of the preceding steps (S1, S2, S3) or step 4 itself is inadmissible — without treating the valid steps as authorization for step 4.

Step 5 is not modeled in the circuit. It is post-execution and its validity or invalidity does not bear on whether step 4 executes. This boundary is stated explicitly in Section 14.

2. Primary Hypothesis

H1 (Primary): In a sequential 5-step workflow where step 4 is the irreversible execution step, the control model blocks step 4 execution when any one or more steps (S1 through S4) is inadmissible — and allows step 4 execution only when all steps S1 through S4 are independently authorized.

Operationally:

- **ALLOW:** $c_{s1} = 1 \text{ AND } c_{s2} = 1 \text{ AND } c_{s3} = 1 \text{ AND } c_{s4} = 1 \rightarrow \text{X gate on } Q_P (Q_P \rightarrow |1\rangle) \rightarrow S_A \geq 0.90$
- **DENY:** any $c_{sN} = 0 (N \in \{1,2,3,4\}) \rightarrow \text{no X gate } (Q_P \text{ stays } |0\rangle) \rightarrow L_D \leq 0.02$

H1 is confirmed if and only if **all four primary pass criteria** in Section 9 are met simultaneously.

3. Secondary Hypotheses

H2a — Step-position universality: The position of the invalid step is irrelevant to the DENY outcome. Whether step 1, 2, 3, or 4 is invalid — and whether one or multiple steps are invalid — the irreversible execution step is blocked in every case. No individual DENY arm leaks above $L_D \leq 0.02$.

H2b — No inherited authorization: Three or more valid prior steps do not authorize the subsequent execution step. Specifically: $c_{s1} = 1, c_{s2} = 1, c_{s3} = 1, c_{s4} = 0$ produces DENY ($L_D \leq 0.02$). The execution step cannot inherit authorization from the approval step.

H2c — Re-authorization restores execution: After a workflow that would have been blocked, a fully re-authorized workflow (all steps valid) executes normally ($S_A \geq 0.90$). The control model is not a one-way block; legitimate complete authorization works.

H2d — Skip-approval attack fails closed: An attempt to skip the approval step ($S3 = 0$) while presenting a valid execution step authorization ($S4 = 1$) fails closed ($L_D \leq 0.02$). Authorization for the execution step cannot substitute for the missing approval step.

H2e — Boundary separation: $\Delta_B = S_A_{\min} - L_D_{\max} \geq 0.70$. The ALLOW and DENY outcomes are metrologically separable with a margin sufficient to be commercially meaningful.

4. Circuit Architecture

4.1 Qubits

- **Q_P** — Payload qubit. Represents the irreversible execution step ($S4$). An X gate ($Q_P \rightarrow |1\rangle$) is applied if and only if all four step authorizations are valid ($c_{\text{exec}} = 1$).

One qubit total. No authorizer qubit is needed: step authorizations are classical constants, not measured quantum outcomes. This is honest to the experiment's scope — see Section 14.

4.2 Classical Step Authorization Model

Each step's authorization status ($c_{s1}, c_{s2}, c_{s3}, c_{s4}$) is a **classical constant** fixed per arm in the circuit code before lock. These are not measured from qubits.

This models the control-logic question: given that the authorization status of each step is definitively known, does the execution gate enforce the AND condition correctly? The distributed-systems question of how step authorizations are obtained in a live deployment is out of scope — see Section 14.

$$c_{\text{exec}} = c_{s1} \text{ AND } c_{s2} \text{ AND } c_{s3} \text{ AND } c_{s4}$$

4.3 Execution Logic

```

Step 1: Prepare Q_P in  $|0\rangle$ 

Step 2: Evaluate step authorizations (classical, per-arm constants)
        $c\_exec \leftarrow c\_s1 \text{ AND } c\_s2 \text{ AND } c\_s3 \text{ AND } c\_s4$ 

Step 3: Conditional execution of irreversible step (S4)
       if  $c\_exec == 1$ :
           Apply X gate to Q_P    [execution – payload driven to  $|1\rangle$ ]
       else:
           Q_P remains in  $|0\rangle$     [halted – irreversible step blocked]

Step 4: Measure Q_P  $\rightarrow$  outcome bit

```

4.4 Implementation Note — Avoiding Nested Conditionals

ARK-444's pre-data correction documented that nested `if_test` blocks with multi-register conditions produce IBM hardware error code 1524 on some backends. ARK-443 resolved this by using sequential single-register conditions. ARK-452 resolves it by a different method: since all four step authorization bits are classical constants (not quantum measurement outcomes), `c_exec` is computed in Python before the circuit is built. The gate structure is then fixed at build time — the X gate is emitted into the circuit if and only if `c_exec == 1`, with no runtime `if_test` conditional at all. This eliminates nested conditionals entirely (and the associated error 1524 risk) while preserving full semantic correctness: the per-arm authorization decision is baked into the compiled circuit deterministically.

4.5 Outcome Interpretation

Q_P Measurement	Interpretation
$P(Q_P = 1) \approx 1.00$	ALLOW — execution gate applied (X); S4 executed
$P(Q_P = 1) \approx 0.00$	DENY — execution gate blocked; workflow halted before S4

5. Arm Specifications

Nine arms. 8192 shots per arm. All nine submitted in a single principal job.

Arm	Label	c_s1	c_s2	c_s3	c_s4	c_exe c	Scen- ario	Ex- pec- ted	Cri- teria
1	AL- LOW- com- plete	1	1	1	1	1	All steps valid; full work- flow au- thor- ized	$S_A \geq 0.90$	H1, H2c, H2e
2	DENY- s1-in- valid	0	1	1	1	0	Read- data step inad- miss- ible; rest valid	$L_D \leq 0.02$	H1, H2a
3	DENY- s2-in- valid	1	0	1	1	0	Calcu- lation step inad- miss- ible; rest valid	$L_D \leq 0.02$	H1, H2a
4	DENY- s3-in- valid	1	1	0	1	0	Ap- proval step inad- miss- ible; execu- tion step other- wise valid	$L_D \leq 0.02$	H1, H2a, H2d
5		1	1	1	0	0	Execu- tion step	$L_D \leq 0.02$	

Arm	Label	c_s1	c_s2	c_s3	c_s4	c_exe c	Scen- ario	Ex- pec- ted	Cri- teria
	DENY- s4-in- valid						itself inad- miss- ible; all prior steps valid		H1, H2a, H2b
6	DENY- s2s3- both	1	0	0	1	0	Two middle steps inval- id; ex- ecu- tion step au- thoriz- ation valid	$L_D \leq 0.02$	H1, H2a
7	DENY- blanke t-at- tempt	1	1	1	0	0	Three valid prior steps; execu- tion step not in- de- pen- dently au- thor- ized — blanke t-auth test	$L_D \leq 0.02$	H1, H2b
8	AL- LOW- reau- th-	1	1	1	1	1	Full re- au- thoriz- ation	$S_A \geq 0.90$	H1, H2c

Arm	Label	c_s1	c_s2	c_s3	c_s4	c_exe c	Scen- ario	Ex- pec- ted	Cri- teria
	com- plete						after prior work- flow block; all steps inde- pen- dently veri- fied		
9	DENY- skip- ap- proval	1	1	0	0	0	Ap- proval and execu- tion steps both ab- sent; data and calcu- lation valid	$L_D \leq 0.02$	H1, H2a, H2d

5.1 Note on Circuit-Equivalent Arms and Scientific Value

Arms 5 and 7 are circuit-equivalent ($c_{s4} = 0$ with $c_{s1} = c_{s2} = c_{s3} = 1$ in both). They are semantically distinct and both necessary:

- **Arm 5 (DENY-s4-invalid):** Tests that the execution step itself can be individually blocked even when all preceding steps are valid. This tests step-level granularity of authorization.
- **Arm 7 (DENY-blanket-attempt):** Explicitly models the “prior valid steps = blanket authorization” failure pattern. The semantic framing is different: here the intent is to use the three prior passes as a substitute for independently authorizing step 4. H2b requires this arm to fail closed, confirming that the control model prevents inherited authorization.

Similarly, Arms 1 and 8 are circuit-equivalent. Arm 1 is the baseline complete workflow. Arm 8 represents a re-authorized workflow after a prior block — validating H2c that the control model is not a one-way gate.

This pattern is consistent with ARK-449's approach and ARK-444's precedent.

6. SPAM Gate

6.1 SPAM Job Specification

Single payload qubit only. The SPAM gate characterizes Q_P readout quality.

- **Shots:** 2048
- **SPAM_P circuit:** Prepare Q_P in $|+\rangle$ via H gate $\rightarrow$ measure $\rightarrow$ $\text{SPAM_P} = |P(Q_P = 1) - 0.5|$

No authorizer qubit is used in ARK-452, so SPAM_A is not applicable. The SPAM gate for this experiment is a single-circuit check on Q_P only.

6.2 SPAM Pass Criterion

Check	Threshold	Pass Condition
SPAM_P	≤ 0.02	Payload

If $\text{SPAM_P} > 0.02$: experiment is **ABORTED AT SPAM GATE**. Principal job results are not read. No threshold loosening, no re-running until it passes.

6.3 SPAM_P Role

SPAM_P is a **gating diagnostic only**. It is not subtracted from DENY leakage. This rule is fixed at lock per the ARK-447 v1.1 correction.

7. Qubit Selection

Select the single qubit Q_P with the lowest readout error from the calibration snapshot taken at submission time. Readout error must be ≤ 0.02 . If no qubit satisfies this constraint, select the lowest-error qubit available and record the deviation explicitly.

Because this experiment uses only one qubit with no inter-qubit gates and no mid-circuit feedforward from a second qubit, connectivity is not a constraint. Select purely on readout quality.

Record at selection time: Q_P index, readout error, calibration snapshot timestamp. Commit before SPAM gate submission.

8. Shot Counts and Execution Plan

Job	Circuits	Shots/Circuit	Total Shots
SPAM gate	1 (Q_P $ +\rangle$)	2048	2048
Principal	9 arms	8192	73,728

Principal job submitted only if SPAM gate passes.

8.1 Strict Execution Sequence

1. Commit MANIFEST (SHA-256 hashes of this document + all circuit files)
→ push → tag ark-452-v1.0-lock
2. Run qubit selection per Section 7
→ record Q_P, readout error in execution log
→ commit execution log entry
3. Submit SPAM gate job
→ record SPAM job ID in execution log
→ commit before reading any results
4. Submit principal job
→ record principal job ID in execution log
→ commit before reading SPAM results
5. Read SPAM gate results
→ apply gate decision per Section 6.2
- 6a. If SPAM gate PASSED → read principal results → apply analysis per Section 11
- 6b. If SPAM gate FAILED → record abort → stop. No principal data read or claimed.

9. Pass / Fail Criteria (Primary — Preregistered)

All four criteria must be satisfied simultaneously for a PASS verdict.

#	Criterion	Threshold	Arms
C1	S_A_min — minimum ALLOW retention across ALLOW arms	≥ 0.90	Arms 1, 8
C2	L_D_max — maximum DENY leakage across all DENY arms	≤ 0.02	Arms 2, 3, 4, 5, 6, 7, 9
C3	$\Delta_B = S_A_{\min} - L_D_{\max}$ — boundary separation	≥ 0.70	Derived
C4	SPAM gate — SPAM_P ≤ 0.02	≤ 0.02	Pre-principal

Failure of any single criterion = FAIL for the entire experiment, regardless of how other criteria performed. A FAIL that identifies which arm or which invalid-step position caused the breach is worth more to the program than a marginal pass that obscures the finding.

10. Secondary Metrics (Descriptive — Not Pass/Fail)

Metric	Description	Hypothesis
S_A per arm	ALLOW retention for Arms 1 and 8 individually	H2c
L_D per arm	DENY leakage for each of Arms 2–7, 9 individually	H2a
L_D by step position	Leakage grouped by which step(s) are invalid	H2a
L_D blanket (Arm 7)	Leakage specifically for the blanket-auth attempt	H2b
L_D skip-approval (Arm 9 vs 4)	Comparison of skip-approval vs approval-only-invalid leakage	H2d
In-situ SPAM drift		$P(Q_P=1)$ at ideal-DENY arms – 0.0

11. Analysis Plan

11.1 Raw Count Extraction

For each arm:

- `counts_0` — Q_P measured as $|0\rangle$
- `counts_1` — Q_P measured as $|1\rangle$
- $P(Q_P = 1) = \text{counts}_1 / (\text{counts}_0 + \text{counts}_1)$

11.2 SPAM Correction

ARK-452 uses one qubit only. No authorizer qubit SPAM_A correction is needed. SPAM_P is the gate diagnostic; it is not subtracted from L_D.

For ALLOW arms: no SPAM correction is applied (no authorizer qubit to correct). $S_A = P(Q_P = 1)$ raw.

For DENY arms: $L_D = P(Q_P = 1)$ raw.

11.3 Primary Metric Computation

```
# ALLOW arms
S_A_arm1 = P1(arm=1)
S_A_arm8 = P1(arm=8)
S_A_min = min(S_A_arm1, S_A_arm8)

# DENY arms
L_D_arms = [P1(arm) for arm in [2, 3, 4, 5, 6, 7, 9]]
L_D_max = max(L_D_arms)

# Boundary separation
Delta_B = S_A_min - L_D_max
```

11.4 Verdict Assignment

```
if not SPAM_gate_passed:
    VERDICT = "ABORTED AT SPAM GATE"
elif S_A_min >= 0.90 and L_D_max <= 0.02 and Delta_B >= 0.70:
    VERDICT = "PASS"
else:
    VERDICT = "FAIL"
```

11.5 Secondary Analysis

- Report L_D per arm to evaluate H2a (step-position universality)
- Report L_D for Arm 5 and Arm 7 to evaluate H2b (no inherited authorization)
- Report L_D for Arm 4 and Arm 9 to evaluate H2d (skip-approval attack)
- Group DENY leakage by number of invalid steps (one invalid vs. two invalid) for descriptive analysis
- If FAIL: identify which criterion and which arm(s) contributed

12. What Constitutes Failure (and Why Each Is Still Valuable)

12.1 L_D > 0.02 for any DENY arm

A specific step-invalidity pattern leaks past the execution gate. The secondary analysis names which arm and what the leak rate is. If, for example, Arm 7 (blanket-auth attempt) leaks while Arms 2-5 hold, this is a specific finding: the control logic blocks single-step invalidity but not the blanket-authorization bypass pattern. This is commercially important to name precisely.

12.2 Arm 7 specifically leaks (H2b)

If valid prior steps do bleed into authorization for step 4, this is the most important failure mode in the experiment. It would directly identify the inherited-authorization vulnerability in the current control model and drive an architectural fix before the result is ever stated commercially.

12.3 SPAM gate fails

Readout condition is outside tolerance at submission time. Experiment aborts. Budget is conserved. New locked run required.

13. Pre-Execution Commitments (Lock Rules)

The following are unconditionally prohibited after the MANIFEST is committed:

1. No criterion changes after lock.
 2. No arm additions or removals after lock.
 3. No rescue-after-failure if SPAM gate fails.
 4. No post-hoc subgroup selection.
 5. SPAM_P is a gate diagnostic only — never subtracted from L_D.
 6. Qubit fixed at SPAM gate submission.
 7. Analysis code committed at lock — runs unmodified on hardware data.
-

14. Honest Boundary Statements

1. Classical step authorization model. Step authorizations (c_s1 through c_s4) are deterministic classical constants set per arm — not obtained from a live authorization service, not communicated between distributed agents, and not subject to race conditions or propagation delays. The experiment proves the execution-gate control logic; it does not model the full complexity of multi-agent step orchestration.

2. Sequential rather than concurrent. The workflow is modeled as a strictly sequential chain. Concurrent step evaluation, parallel workflow branches, or competing workflow instances are out of scope.

3. Step 5 not modeled. The write-record step (S5) is post-execution. Its validity or invalidity does not affect whether S4 executes. The experiment makes no claim about post-execution step governance.

4. Single irreversible step. The workflow has exactly one irreversible step (S4). Workflows with multiple irreversible steps, or with reversible-then-irreversible sequencing within a single step, are more complex and deferred.

5. Hardware scope. Results apply to the specific backend, qubit, calibration snapshot, and shot counts used.

6. Not a cryptographic proof. These are hardware noise-characterization studies under a preregistered protocol.

15. Connection to Corpus and Forward Roadmap

15.1 What ARK-452 Adds (if PASS)

The corpus would then span: authorization boundary · expiry · tamper · quorum · three-state · noise · gate-stop · state-change currency (ARK-449) · **multi-step workflow governance** (ARK-452).

This is the transition point from “isolated boundary enforcement” to “agentic workflow governance” — the language that connects ExecutionProof directly to the AI agent safety commercial market.

15.2 What ARK-452 Enables Downstream

- **ARK-451 (Mid-Execution Revocation):** ARK-452 establishes per-step authorization in a sequential chain. ARK-451 extends that to revocation of authority during an active execution window — after step 3 has passed but before step 4 completes.
- **ARK-455 (ProofRecord Tamper Verification):** A ProofRecord from a complete workflow (Arm 1 or 8) contains richer structure than a single-action record. ARK-455 gains a more complex artifact to tamper with.

15.3 Items Explicitly Deferred

- **Concurrent workflow branches** — deferred; requires multi-qubit entanglement across branches, out of scope for this series.
- **Mid-execution step revocation** — ARK-451.
- **Workflow step conflict (one step says proceed, another says halt)** — deferred to ARK-453 framing applied to workflow steps.

16. MANIFEST Placeholder

```
ARK-452 MANIFEST – Version 1.0
Lock timestamp: [FILL AT LOCK – UTC, YYYY-MM-DDTHH:MM:SSZ]
Backend target: ibm_marrakesh (156-qubit Heron r2)

SHA-256 hashes:
  ARK_452_preregistration.md:      [COMPUTE AT LOCK]
  circuits/ark_452_circuit.py:     [COMPUTE AT LOCK]
  circuits/ark_452_analysis.py:    [COMPUTE AT LOCK]

MANIFEST SHA-256:                  [COMPUTE LAST]

Committed to:   executionproof-testbeds
Tag:            ark-452-v1.0-lock
Commit SHA:     [FILL AFTER PUSH]

SPAM job ID:    [FILL BEFORE READING ANY RESULTS]
Principal job ID:[FILL BEFORE READING SPAM RESULTS]
```

17. ProofRecord Stub

```
{
  "experiment": "ARK-452",
  "doctrine_tested": "Prior valid steps in a workflow do not authorize the irrevers-
ible execution step. Each step requires independent authorization.",
  "verdict": "[PASS | FAIL | ABORTED AT SPAM GATE]",
  "timestamp_lock": "[UTC]",
  "timestamp_execution": "[UTC]",
  "backend": "ibm_marrakesh",
  "qubit": {"Q_P": null},
  "workflow_steps": ["S1:read-data", "S2:calculate", "S3:approve-payment", "S4:execute
-payment (irreversible)", "S5:write-record (not modeled)"],
  "spam_gate": {"SPAM_P": null, "passed": null},
  "primary_metrics": {
    "S_A_arm1": null, "S_A_arm8": null, "S_A_min": null,
    "L_D_max": null, "L_D_max_arm": null, "Delta_B": null
  },
  "secondary_metrics": {"L_D_per_arm": {}, "L_D_blanket_arm7": null},
  "criteria": {
    "C1_S_A_min_pass": null, "C2_L_D_max_pass": null,
    "C3_Delta_B_pass": null, "C4_SPAM_pass": null
  },
  "manifest_sha256": "[FILL AT LOCK]",
  "zenodo_doi": "[FILL AT PUBLICATION]"
}
```

This document is the preregistration for ARK-452. It is locked at the time of MANIFEST commit and tag. No changes to hypotheses, criteria, arms, thresholds, or analysis plan are permitted after that point. A FAIL or ABORT is a valid, publishable result that advances the program.

Remnant Fieldworks Inc. — ExecutionProof ARK Series
<https://github.com/derekhone/executionproof-testbeds>